A RELATIVELY DEFINABLE FUNCTION ON THE REALISATIONS OF A SIMPLE TYPE IS A POWER OF THE SUCCESSOR

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Allowed standard prerequisites (editorial).

- The clauses of the basic lemma on simple types, in particular that every relatively definable nonempty subset of p(M) has a minimum. Same source, Lemma labelled Lemma_basic_simple, clause (c); invoked inside the counted proof at lines 224 and 229. Supplies the minimum used for injectivity and the contradiction with non-minimality used for order preservation.
- Aleph_0-saturation of the ambient model and compactness. Used inside the counted proof at lines 223, 231 and 236. Gives surjectivity of the extension of f and lets the interval-wise behaviour found on a tail of C be transferred to the whole type p.

Source reference. Original lemma 1.4: sources/vaught-discrete-orders-2212.13605v3/source0.tex, source lines 224–224. Navigation pointer for \ref{Lemma_basic_simple} used inside the counted span (line 224); the argument that consumes it is retained in the span.

Lemma 0.1. *Suppose that $p \in S_1(T)$ is a simple type, as witnessed by $(C, <)$ and $(D, <)$. If $f : p(M) \to p(M)$ is a relatively definable function, then $f = S^k$ for some $k \in \mathbb{Z}$, where S is the successor function on $(p(M), <)$.*

Proof. Let $c_0 = \min C$, $c_n = S^n(c_0)$ for $n \in \mathbb{N}$, and suppose that $(D, <)$ is discrete. First, we will prove that f is an automorphism of $(p(M), <)$. Choose a definable function $\hat{f} : D_1 \to D_2$ extending f. Since $p(x) \cup \{\hat{f}(x) = a\}$ is consistent for some (equivalently all) $a \in p(M)$ and M is $\aleph_0$-saturated, f is surjective. To verify the injectivity of f, let us consider the set $\hat{f}^{-1}(\{f(a)\}) \cap [c_0, a] \subset p(M)$. It is an a-definable subset of $p(M)$ so, by Lemma 1.4(c), has minimum, say a'; clearly, $a' = \min(\hat{f}^{-1}(\{f(a')\}) \cap [c_0, a'])$. Since $a' \in p(M)$, the formula $x = \min(\hat{f}^{-1}(\{f(x)\}) \cap [c_0, x])$ belongs to $p(x)$; it follows that f is injective. In

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particular, the inverse function f^{-1} is relatively definable, so after possibly shrinking D_1 and D_2, we will assume that $\hat{f}$ is a bijection. It remains to show that f (equivalently f^{-1}) is order-preserving. Note that at least one of the formulae $\hat{f}(x) \leqslant x$ and $\hat{f}^{-1}(x) \leqslant x$ belongs to $p(x)$. We will from now on assume $(\hat{f}(x) \leqslant x) \in p(x)$, the proof in the other case is similar. Suppose, by way of contradiction, that f is not order-preserving. Then the set $D(a)$ defined by $\hat{f}(a) \leqslant \hat{f}(x) < x < a$ is nonempty (for all $a \in p(M)$). Observe that $D(a) \subset [f(a), a] \subset p(M)$, so $\varnothing \neq D(b) \subset p(M)$ holds for all $b \in D(a)$. Choose $b' \in D(b)$. Then:

$$\hat{f}(a) \leqslant \hat{f}(b) \leqslant \hat{f}(b') < b' < b < a$$

witnesses non-minimality of b in $D(a)$. Hence no element of $D(a)$ is minimal, contradicting Lemma 1.4(c). Therefore, f is an automorphism of $(p(M), <)$.

By compactness, we can modify $\hat{f}$ so that it is strictly increasing. In fact, in what follows we will view $\hat{f}$ as a definable, increasing, partial function $\hat{f} \subset D \times D$. Since $\hat{f}$ is increasing and $\hat{f}(p(M)) = p(M)$, $\hat{f}(C) \subseteq C$.

Let $\phi(x, y)$ be a formula implying $x < y$ and expressing "$[x, y] \subset \mathrm{dom}(\hat{f})$ and $\hat{f}\restriction[x, y]$ is an order-isomorphism of $(D, <)$-intervals $[x, y]$ and $[\hat{f}(x), \hat{f}(y)]$". Then:

$$p(x) \cup p(y) \cup \{x < y\} \vdash \phi(x, y).$$

By compactness, there is a formula $\theta(x) \in p(x)$ such that:

$$\models \forall x \forall y(\theta(x) \wedge \theta(y) \wedge x < y \to \phi(x, y)). \tag{1}$$

Since p is a simple type and $\theta(x) \in p$, the set $\neg\theta(C)$ is finite, so there is a $n_0 \in \omega$ such that $\{c_i \mid i \geqslant n_0\} \subset \theta(C)$. For all $n \geqslant n_0$, by (1), f maps the (two-element) interval $[c_n, c_{n+1}]$ onto the interval $[\hat{f}(c_n), \hat{f}(c_{n+1})]$, so $\hat{f}(c_n)$ and $\hat{f}(c_{n+1})$ are consecutive elements of D, i.e $\hat{f}(S(c_n)) = S(\hat{f}(c_n))$ (where S is the succesor function on $(D, <)$). By induction we easily get $\hat{f}(S^m(c_{n_0})) = S^m(\hat{f}(c_{n_0}))$ for all $m \in \omega$. Since $\hat{f}(c_{n_0}) \in C$, there is a $k \in \mathbb{Z}$ with $\hat{f}(c_{n_0}) = S^k(c_{n_0})$. Then for all $n \geqslant n_0$:

$$\hat{f}(c_n) = \hat{f}(S^{n-n_0}(c_{n_0})) = S^{n-n_0}(\hat{f}(c_{n_0})) = S^{n-n_0+k}(c_{n_0}) = S^k(S^{n-n_0}(c_{n_0})) = S^k(c_n).$$

Hence $(\hat{f}(x) = S^k(x)) \in \mathrm{tp}(c_n)$ for all $n \geqslant n_0$. Since p is a C-type, $(\hat{f}(x) = S^k(x)) \in p(x)$. □

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